Yu, Zijun

zijun.yu@mail.mcgill.ca

PhD student in Mathematics and Statistics developing distribution-free methods for uncertainty quantification and risk control, with a current focus on the reliability, evaluation, and safety of LLMs and autonomous agents. Co-first-author papers at ICML 2026 and NeurIPS 2026, plus a NeurIPS workshop paper on agent safety and one manuscript under review.

Education

2025 – 2029 (exp.) **Ph.D. in Mathematics and Statistics** (direct-entry)
McGill University, Montreal, QC
Supervisors: Prof. Masoud Asgharian, Prof. Vahid Partovi Nia
Research area: distribution-free uncertainty quantification and risk control; reliability and safety of LLMs and autonomous agents

2016 – 2020 **B.Sc. Honours in Statistics and Computer Science**
McGill University, Montreal, QC
First-Class Honours Degree

Awards and Scholarships

- **ISM Graduate Scholarship**, Institut des sciences mathématiques 2026–2027
- **ISM Travel Grant**, Institut des sciences mathématiques 2026
- **ISM Graduate Scholarship**, Institut des sciences mathématiques 2025–2026
- **Graduate Excellence Award**, McGill University 2025–2029
- **First-Class Honours Degree**, McGill University 2020
- **Science Undergraduate Research Award (SURA)**, McGill University 2019
- **James McGill Scholarship**, McGill University 2016

Publications

† Equal contribution.

Conference

- Yu Gu†, **Zijun Yu**†, Vahid Partovi Nia, Masoud Asgharian. "Pause and Reflect: Conformal Aggregation for Chain-of-Thought Reasoning." *Conference on Neural Information Processing Systems (NeurIPS)*, 2026. arXiv:2605.14098
- Yu Gu†, **Zijun Yu**†, Chi-Kuang Yeh, Xinyu Wang, Ziyang Song. "Severity-Controlled Prediction Sets for Medication Recommendation." *Conference on Neural Information Processing Systems (NeurIPS)*, 2026.
- Pengqi Liu†, **Zijun Yu**†, Mouloud Belbahri, Arthur Charpentier, Masoud Asgharian, Jesse C. Cresswell. "Beyond Procedure: Substantive Fairness in Conformal Prediction." *International Conference on Machine Learning (ICML)*, 2026. arXiv:2602.16794

Workshop

- **Zijun Yu**†, Yu Gu†, Vahid Partovi Nia, Masoud Asgharian. "G-CARB: Graph-Localized Conformal Agent Risk Budget for Compositional Harm." *NeurIPS Workshop on SLMs for Agentic Systems (SLM-Agents)*, 2026. arXiv:2610.05563

Under Review

- **Zijun Yu**†, Yu Gu†, Vahid Partovi Nia, Masoud Asgharian. "Conformal Prediction under Partial Verification." *Under review*, 2026. arXiv:2610.10829

Presentations

- "Pause and Reflect: Conformal Aggregation for Chain-of-Thought Reasoning." Poster presentation, *Conference on Neural Information Processing Systems (NeurIPS)*, Sydney, Australia, 2026.
- "Severity-Controlled Prediction Sets for Medication Recommendation." Poster presentation, *Conference on Neural Information Processing Systems (NeurIPS)*, Atlanta, USA, 2026.
- "G-CARB: Graph-Localized Conformal Agent Risk Budget for Compositional Harm." Workshop presentation, *NeurIPS Workshop on SLMs for Agentic Systems (SLM-Agents)*, Paris, France, 2026.
- "Beyond Procedure: Substantive Fairness in Conformal Prediction." Poster presentation, *International Conference on Machine Learning (ICML)*, Seoul, South Korea, 2026.

Patents and Open-Source Software

- **Patent application** filed (Layer 6 AI, 2026), related to methods in 'Beyond Procedure: Substantive Fairness in Conformal Prediction.'
- `llm-in-the-loop-conformal-fairness` — open-source release of the evaluation and benchmarking framework for the ICML 2026 paper. github.com/layer6ai-labs/llm-in-the-loop-conformal-fairness

Research and Work Experience

2025 – 2026 *Mitacs Internship — Student Researcher*
Layer 6 AI, Toronto

- Conducted research on substantive fairness in conformal prediction, leading to "Beyond Procedure: Substantive Fairness in Conformal Prediction" (ICML 2026) and a filed patent application.
- Designed and architected a modular codebase supporting large-scale experiment pipelines, enabling reproducible benchmarking across multiple fairness criteria, datasets, and model families; released open source.
- Collaborated with research scientists on theoretical analysis and empirical validation.

2020 – 2024 *Software Developer*
OneDesk Inc., Montreal

- Built a Retrieval-Augmented Generation chatbot pipeline (web crawler, vector database, LLM APIs) and added data analysis and visualization capabilities to the core product.
- Diagnosed and resolved memory leaks, race conditions, and data-consistency failures across frontend and backend systems; applied time-series analysis to production logs to audit system performance and identify regressions.
- Led and mentored a team of junior developers across multiple concurrent projects.

Technical Skills

ML: PyTorch, Hugging Face Transformers, vLLM, llama.cpp
Languages: Python, R, Java, TypeScript, SQL
Infrastructure: Docker, PostgreSQL, Redis, Git, Proxmox VE